Supplementary Information

Coverage and tissue geometry govern the reconstruction of spatial transcriptomics from sparse molecular measurements

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Contents

This Supplementary Information contains three interpretive notes, 12 figures and nine tables. The model definitions and detailed training and evaluation procedures are presented in the main Methods and its Experimental setup subsection. Tables S1 and S2 report the primary gene panel and patient-level reconstruction results. Tables S3–S7 define the metrics, negative controls, numerical verification, observation masks and distance-resolved summaries. Table S8 summarizes the selected supporting architectures, and Table S9 reports reconstruction across measurement budgets and missing-region protocols. Corresponding numerical data are supplied in `source_data/` and `tables/`; the per-gene table is a gzip-compressed CSV.

S1 Supplementary notes

S1.1 Affine interpolation identity

For a gene with transform $T(y) = (y - \mu)/\sigma$, $\sigma > 0$, and interpolation weights satisfying $\sum_j w_{ij} = 1$, we have

$$T^{-1}\left(\sum_j w_{ij} T(y_j)\right) = \sigma \sum_j w_{ij} \frac{y_j - \mu}{\sigma} + \mu = \sum_j w_{ij} y_j.$$

Consequently, applying and inverting a fixed per-gene affine transform does not create a distinct molecular interpolation method when weights are unchanged. In the Wu evaluation, the raw and transformed predictions differed by at most 2.66×10^{-15}, consistent with roundoff. Tiny numerical changes can break tied ranks and alter raw Spearman calculations; such changes were logged as numerical rank sensitivity, not counted as harmonization benefits. Calibration can still alter an image branch or a learned fusion model; that is a different comparison. The manuscript therefore omits duplicate molecular interpolation curves and makes no claim of an affine-harmonization advantage for molecular interpolation itself.

S1.2 Dataset roles, masking and interpretation

The four-patient HEST cohort was used for development. HER2ST was first evaluated under a frozen protocol and subsequently used for calibration development; those two views reuse the same eight patients. The Wu cohort supplied three calibration and three evaluation patients. The 22 Bassiouni patients were evaluated without new cohort-specific fitting. Counts of independent evaluation patients must not add repeated HER2ST analyses or mask replicates as new patients.

Distributed evaluation reveals molecular vectors at sampled anchors and scores all remaining locations. The external regional-completion protocol instead withholds a contiguous 40% region, samples anchors outside it, and scores that region only. V1 uses inverse-distance-squared interpolation and V2 uses an equal-weight nearest-anchor mean. Their contrast is therefore a protocol-level difference, not an isolated geometry intervention. The earlier development block/capture masks are also not identical to the external forbidden-region design.

Public external H&E images are downsampled, and sparse molecular interpolation receives same-section expression unavailable to the image-only baseline. Image-resolution limitations and unequal available information must accompany comparisons with histology. An H&E-fusion failure under these frozen inputs does not establish that native-resolution histology is generally uninformative.

S1.3 Aggregation and metric interpretation

The primary external summaries compute gene-level metrics on hidden locations, summarize genes within each run, average random seeds within geometry, average geometries within patient, and finally summarize patients. Unless a figure legend states otherwise, external cohort markers are medians across patients. Several development and transportability plots instead show arithmetic means; their legends state this explicitly. A difference between two cohort medians is not a median paired patient effect.

Mean squared error and mean absolute error use the unchanged development-standardized log1p-count target. Shared numerical units do not remove capture-depth or assay differences. Correlations are undefined when the target or prediction range is at most 10^{-12}; they are excluded from correlation averages but not counted as successes in threshold fractions. Spearman correlations use average ranks for ties. Mask and model seeds describe computational variation, not independent biological replication. No confidence intervals or significance tests have been added.

S2 Supplementary figures

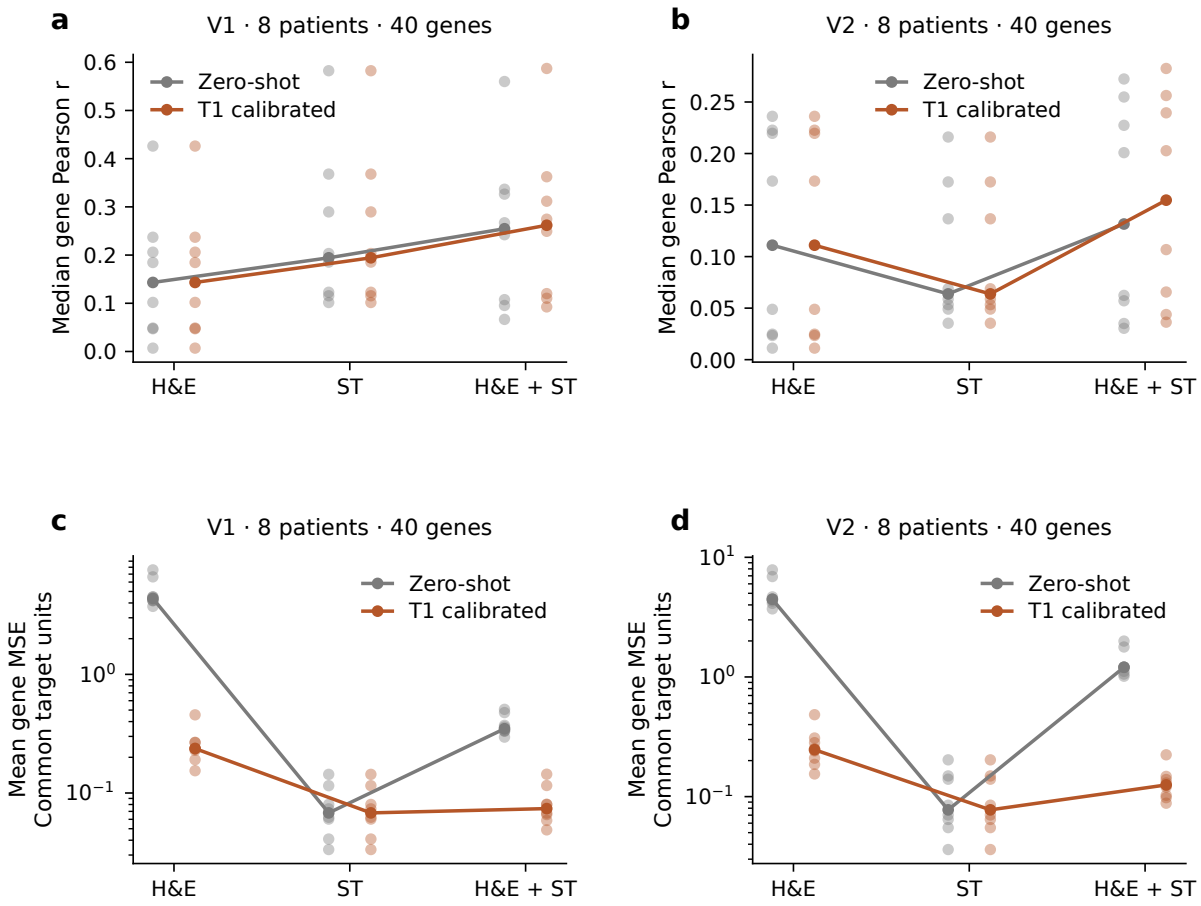


Figure S1: HER2ST transportability and target calibration. Eight HER2ST patients and the frozen 40-gene intersection are shown at 20% random measured-location coverage. **a,c,** Distributed interpolation (V1). **b,d,** Contiguous regional completion (V2). Panels a/b show median gene Pearson correlation and c/d show mean gene MSE in common target units on a logarithmic axis. Dots are patient summaries after averaging seeds and geometries; connected markers are medians across patients. T0 denotes the uncalibrated zero-shot analysis, whereas T1 denotes the calibrated target mapping with fixed fusion. Image weights were 0.25 for V1 and 0.5 for V2; these development weights differ from the final external weight of 0.2. These are the same patients, not independent replication groups. The T1 fixed-fusion analysis differs from the later fold-selected fusion results in Supplementary Fig. S10. Affine transformation and inverse transformation leave molecular interpolation unchanged; apparent image/fusion calibration effects are not a new interpolation advantage.

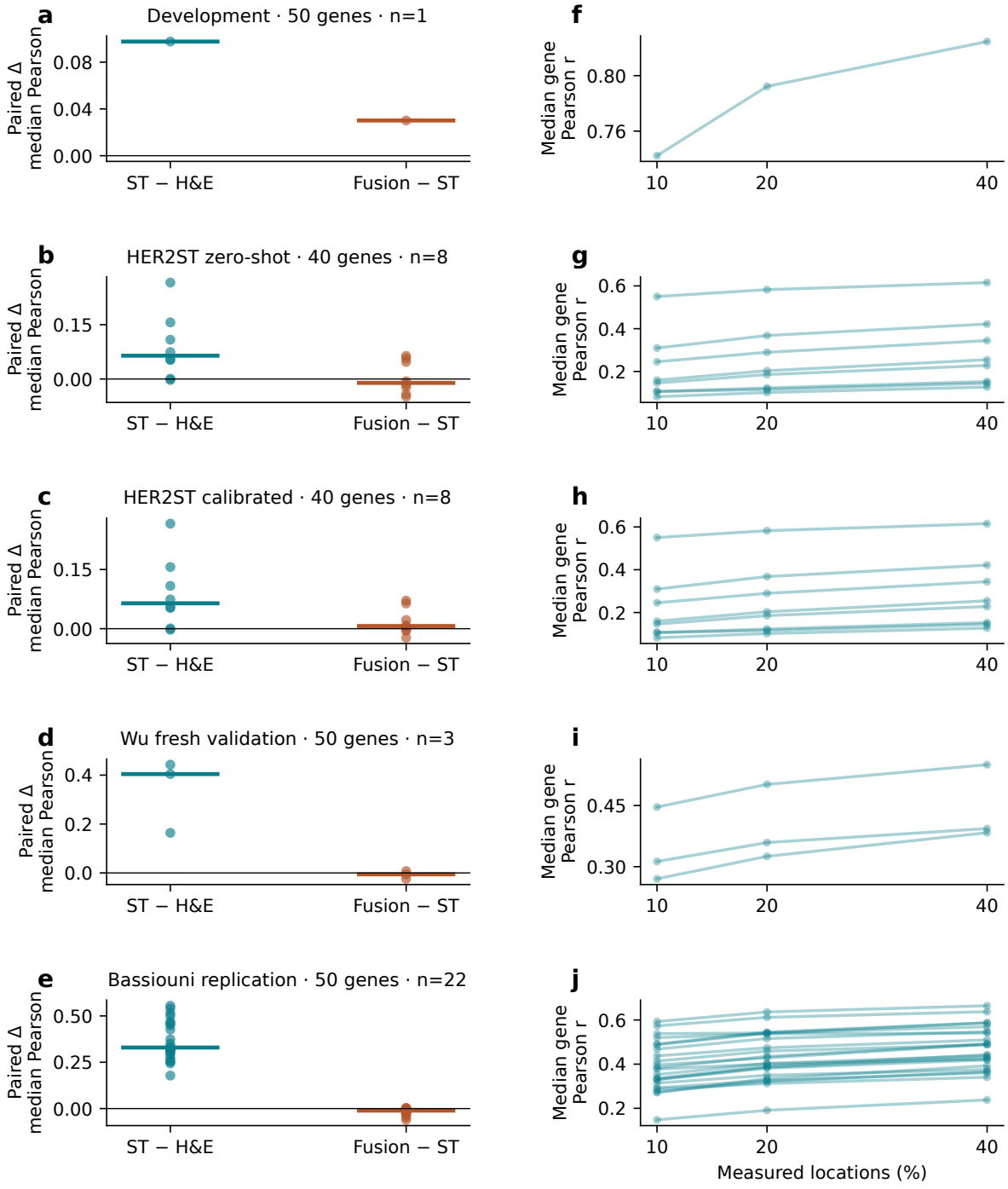


Figure S2: Cross-cohort synthesis without pooling different panels. Rows correspond to the development primary patient ($n = 1$), HER2ST zero-shot ($n = 8$), HER2ST calibrated ($n = 8$, the same patients), fresh Wu evaluation ($n = 3$) and fresh Bassiouni replication ($n = 22$). The development and fresh cohorts use 50 genes; HER2ST uses 40. **a–e,** Left-column paired patient correlation effects at 20% distributed coverage for molecular interpolation minus image-only prediction and fusion minus molecular interpolation; bars are median paired effects. **f–j,** Right-column individual patient molecular correlation curves at 10%, 20% and 40% distributed coverage. The panels preserve differences in analytical role and gene panel and do not define a pooled cross-cohort score. HER2ST calibration is development, not an additional fresh validation cohort.

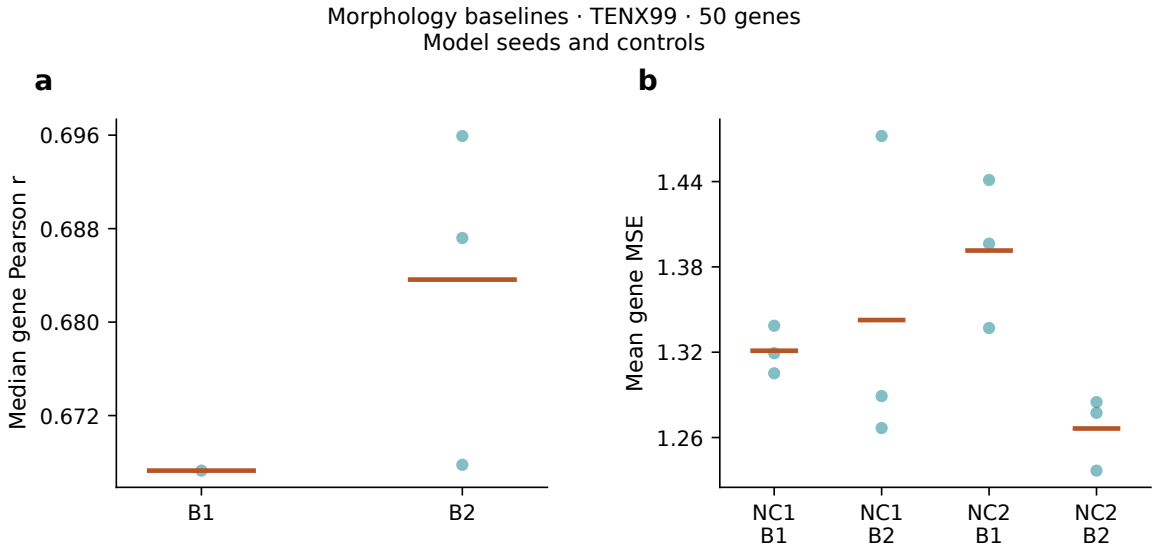


Figure S3: Morphology-only development baselines and controls. The morphology-only development analyses use the single primary development patient TENX99 and 50 genes. **a,** Median gene Pearson correlation for ridge regression (B1) and the small multilayer perceptron (B2). The training-mean baseline has constant predictions and undefined correlations and is not drawn as zero. **b,** MSE under training-target-row permutation (NC1) and training-embedding-row permutation (NC2). Dots represent saved model-seed runs and bars are arithmetic means of displayed runs, not independent patient estimates. The controls disrupt image–expression pairing while preserving validation and evaluation data. This initial image-only head differs from the frozen local MLP selected in the representation comparison and used for external replication.

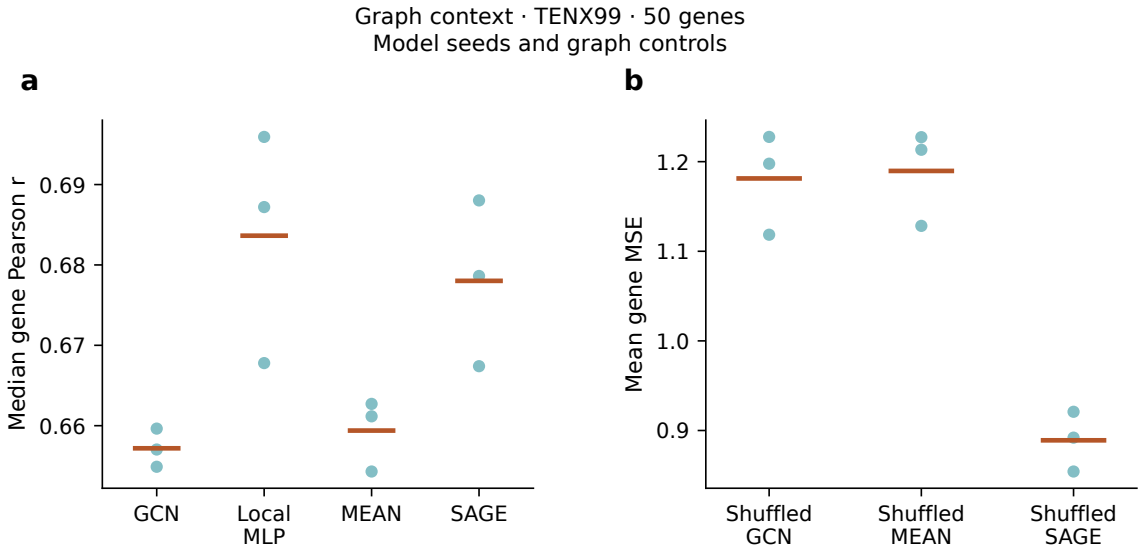


Figure S4: Single-scale graph context did not by itself establish spatial specificity. The single-scale graph comparison uses TENX99 and the 50-gene panel. **a,** Median gene correlation for the frozen local MLP, neighborhood mean pooling plus MLP, GCN and GraphSAGE. **b,** Mean gene MSE for the archived shuffled-neighborhood runs. Dots are training-seed runs and bars are their arithmetic means. A graph model outperforming a local predictor does not establish that the correct spatial adjacency caused the gain; generic aggregation or model parameterization can also contribute. The results use the original fitted models without additional graph selection or local-predictor refitting.

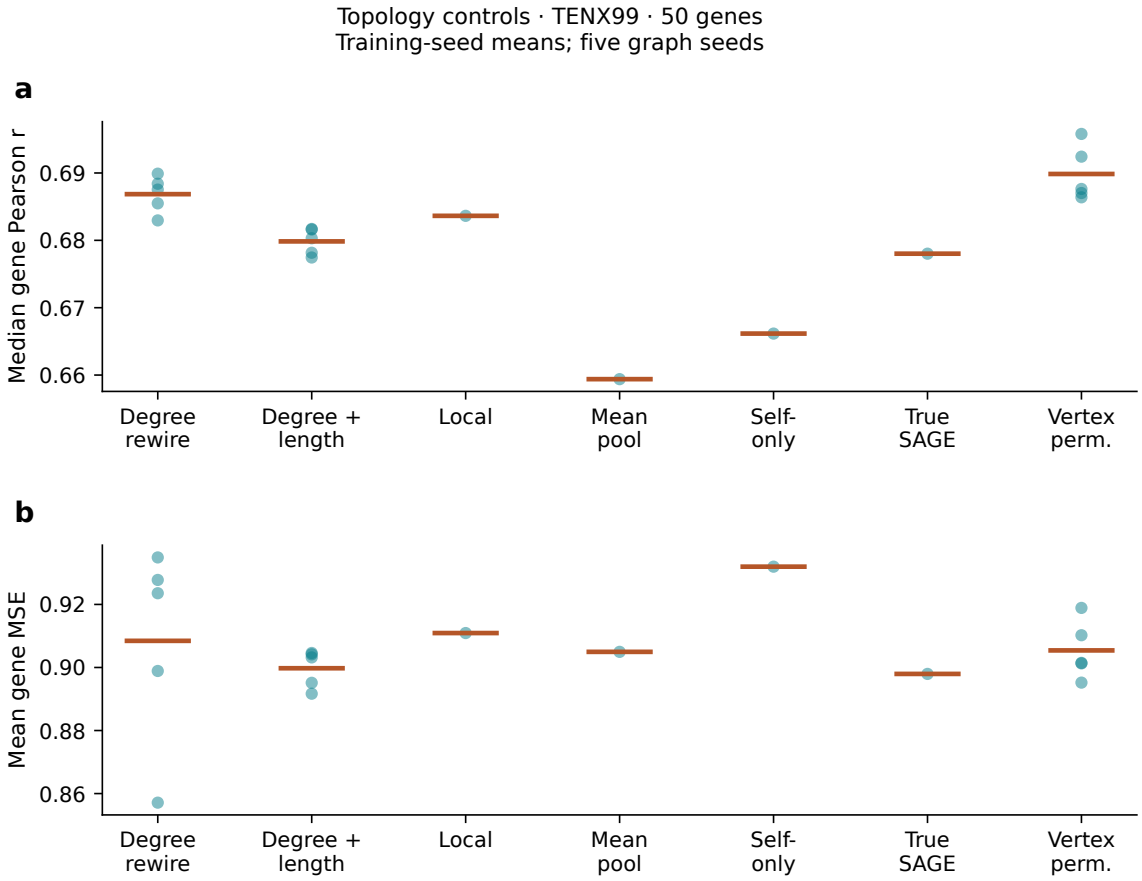


Figure S5: Matched graph controls limited claims about exact topology. The matched graph controls use one development patient, TENX99, and 50 genes. **a,** Median gene Pearson correlation. **b,** Mean gene MSE. Local, self-only, mean-pooling and true-spatial GraphSAGE results accompany vertex-permuted, degree-rewired and degree-plus-edge-length-rewired controls. Training seeds are averaged within each graph randomization seed; the randomized controls have five graph seeds. Dots are these averages and bars are arithmetic means. Random graph seeds are not biological replicates. The experiment did not establish a reproducible advantage of the true spatial topology. This figure records a negative/mixed development result, not evidence that topology is irrelevant in every representation or tissue.

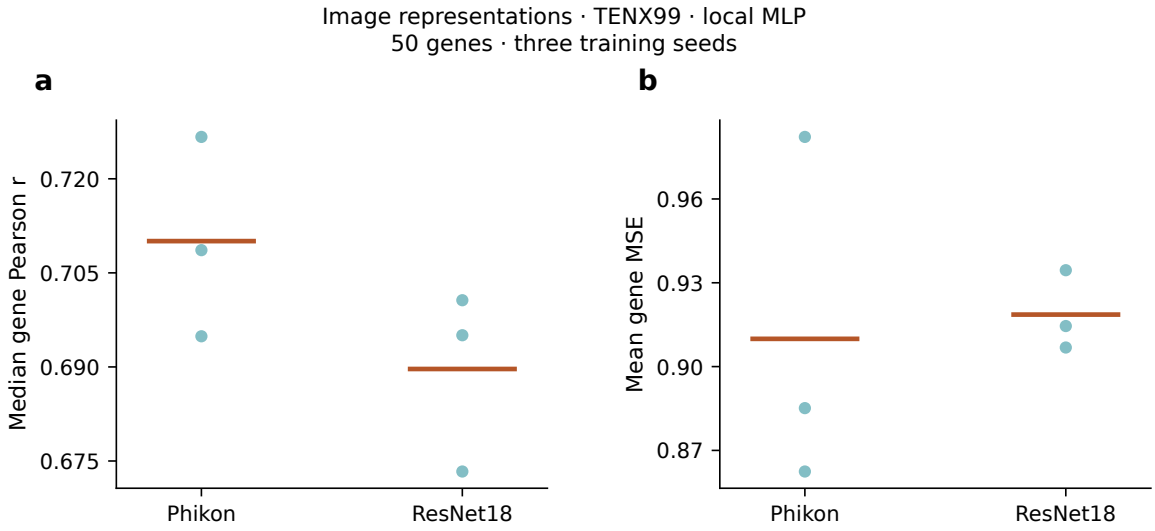


Figure S6: Pathology representation effects in the development patient. The local MLP representation comparison evaluates frozen ImageNet ResNet18 and Phikon representations in TENX99 for 50 genes. **a,** Median gene Pearson correlation. **b,** Mean gene MSE. Each dot is one of three training seeds; bars are arithmetic means. The panel-specific improvement did not establish general superiority on larger target panels or reproducible topology-specific gains. Phikon is not introduced into the final Bassiouni replication; that analysis retains the frozen ResNet18 image branch. These are development comparisons, not external encoder validation.

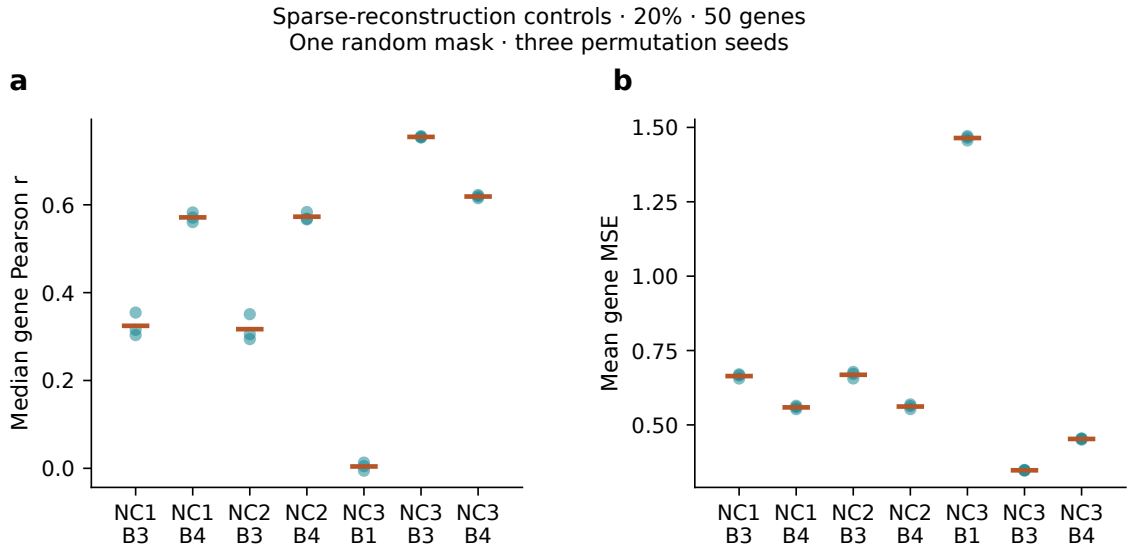


Figure S7: Sparse-reconstruction development controls. The sparse-reconstruction controls use the ResNet18 branch, TENX99, 50 genes and 20% measured-location coverage. In this figure, B1 denotes image-only prediction, B3 image–molecular fusion and B4 learned propagation. Controls include anchor-expression permutation (NC1), anchor-position permutation (NC2) and image-location permutation (NC3). Model labels are specific to this experiment. **a,** Median gene Pearson correlation. **b,** Mean gene MSE. All controls use the same random observation mask (replicate 0); dots represent three permutation seeds and bars are arithmetic means. These distributions are control diagnostics rather than estimates of a geometry effect. Separate integrity checks establish invariance to corruption of hidden labels. Related anchor-assignment disruptions must not be counted as independent biological mechanisms.

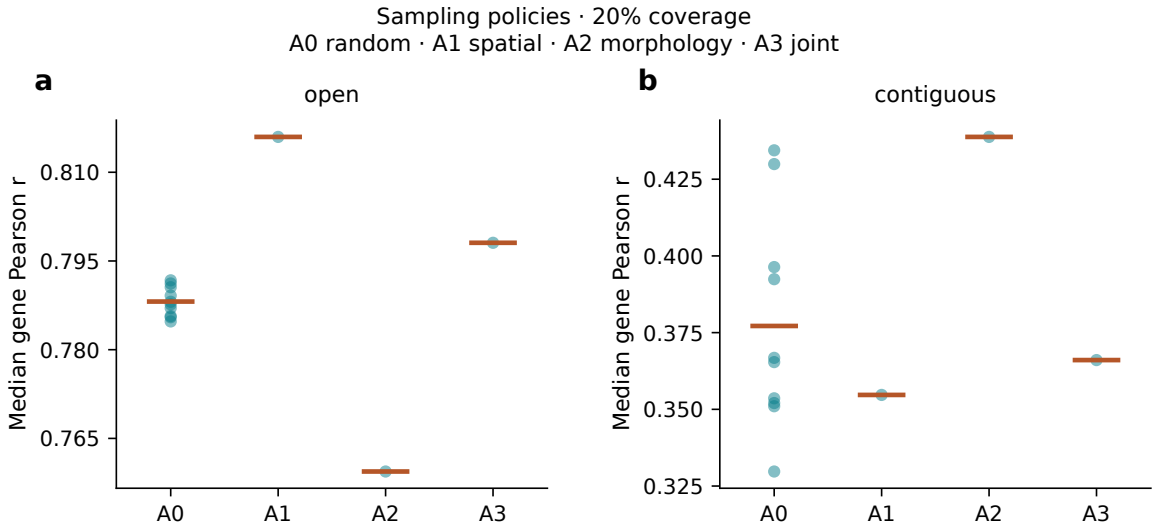


Figure S8: Sampling effects depended on the reconstruction regime. The sampling comparison uses interpolation-only results for TENX99, the ResNet18 reference branch, 50 genes and 20% coverage. **a,** Open-tissue interpolation. **b,** Contiguous missing-region evaluation. A0 is random sampling with ten seeds; A1 is spatial farthest-point sampling, A2 morphology-space farthest-point sampling, and A3 a locked joint policy. A1–A3 are deterministic given their stored initialization. Dots are policy runs and bars are arithmetic means. This limited baseline diagnostic is not a complete summary of all learned/fusion families and does not demonstrate universally optimal sampling.

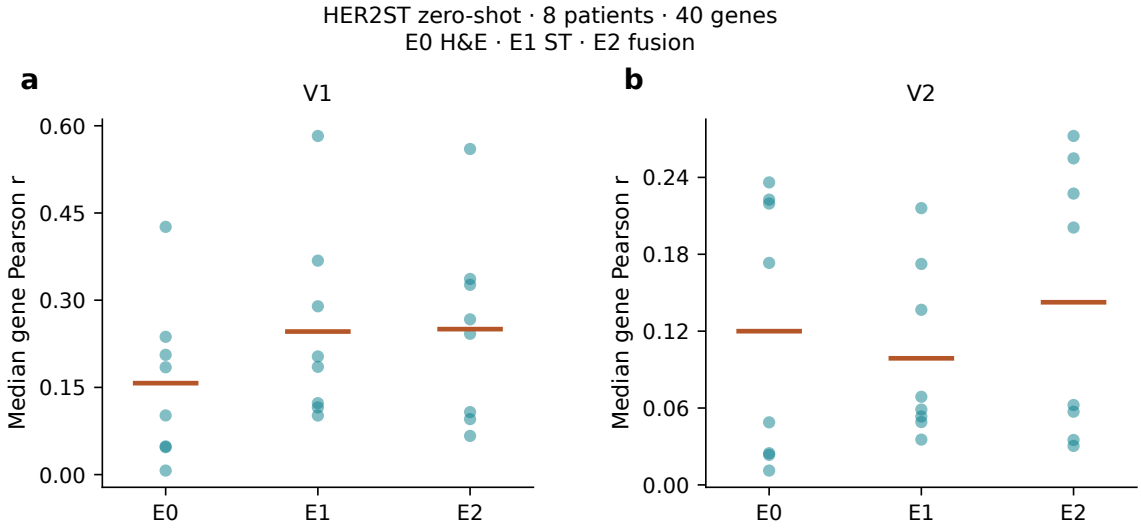


Figure S9: HER2ST patient distributions under the frozen zero-shot protocol. The eight HER2ST patients, 40 shared primary genes and 20% random-anchor coverage are summarized separately for **a,** distributed interpolation and **b,** contiguous regional completion. E0 denotes H&E-only prediction, E1 molecular interpolation and E2 frozen fusion, with image weights 0.25 for V1 and 0.5 for V2. Dots are patient means across seeds/geometries. Bars are arithmetic patient means, unlike the median summaries in Supplementary Fig. S1 and the primary tables. These zero-shot results and the subsequent calibration analyses use the same patients; HER2ST therefore does not supply an untouched final validation cohort.

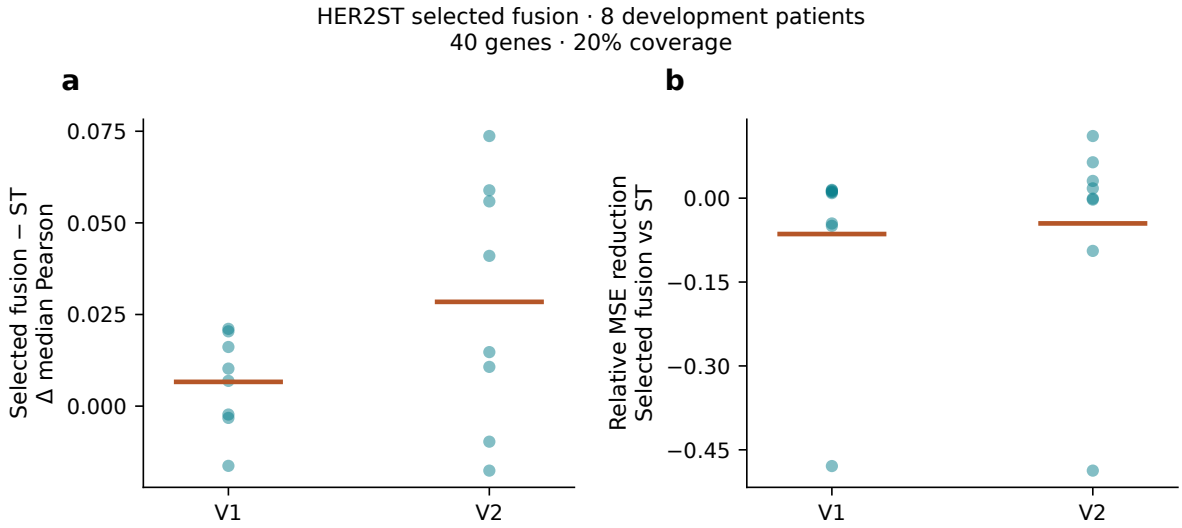


Figure S10: Patient-disjoint calibration did not establish robust fusion superiority. The calibration-development analysis includes eight HER2ST patients, the frozen 40-gene intersection and 20% coverage. **a,** Selected calibrated fusion minus molecular interpolation in Pearson correlation. **b,** Relative MSE reduction compared with molecular interpolation; positive values indicate lower fusion error. Dots are patient/fold results grouped by V1/V2 and bars are arithmetic means. The selected fusion rule was determined within patient-disjoint training/calibration folds, not from the displayed outer-patient outcomes. Joint correlation and error improvements were not consistent across patients. These selected-fusion results should not be interchanged with the T1 fixed-fusion target study shown in Supplementary Fig. S1.

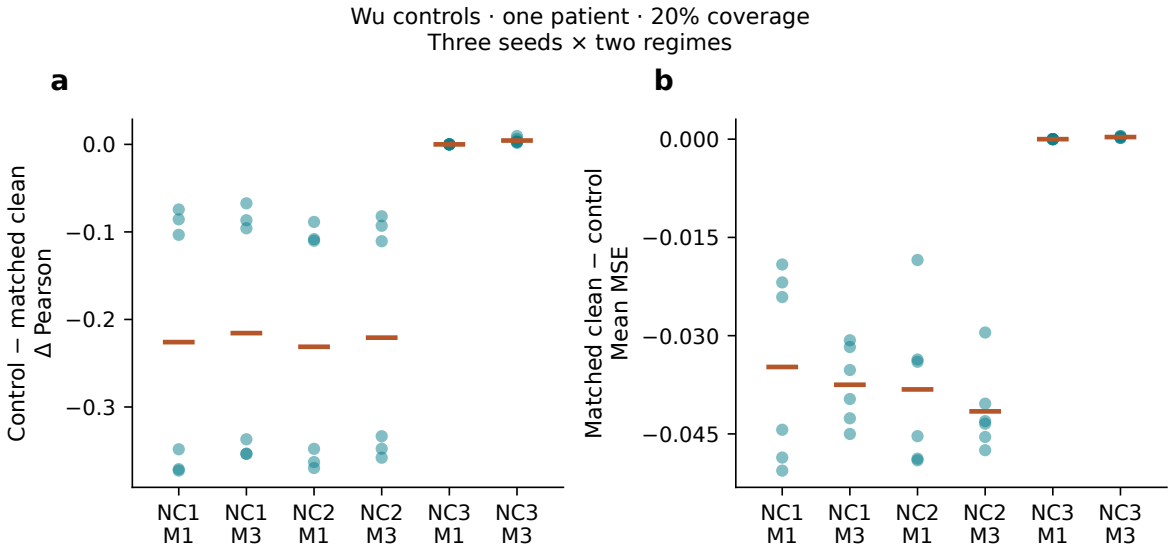


Figure S11: Matched negative controls in fresh Wu evaluation. The predeclared held-out patient 1160920F, 50 genes, 20% coverage and random mask seed 0 were used, with V1 geometry 0 and V2 geometry 1. Three permutation seeds were evaluated for NC1 anchor-vector reassignment, NC2 anchor-position reassignment and NC3 image-prediction permutation. **a,** Control minus matched-clean median gene Pearson correlation. **b,** Matched-clean minus control MSE, so negative values indicate increased control error. Molecular interpolation (M1) and fixed image fusion (M3) are shown. Dots are matched control settings and bars are means; they are not independent patients. Molecular controls disrupt the correct spatial anchor assignments, whereas image permutation provides no robust supporting evidence for the frozen image contribution.

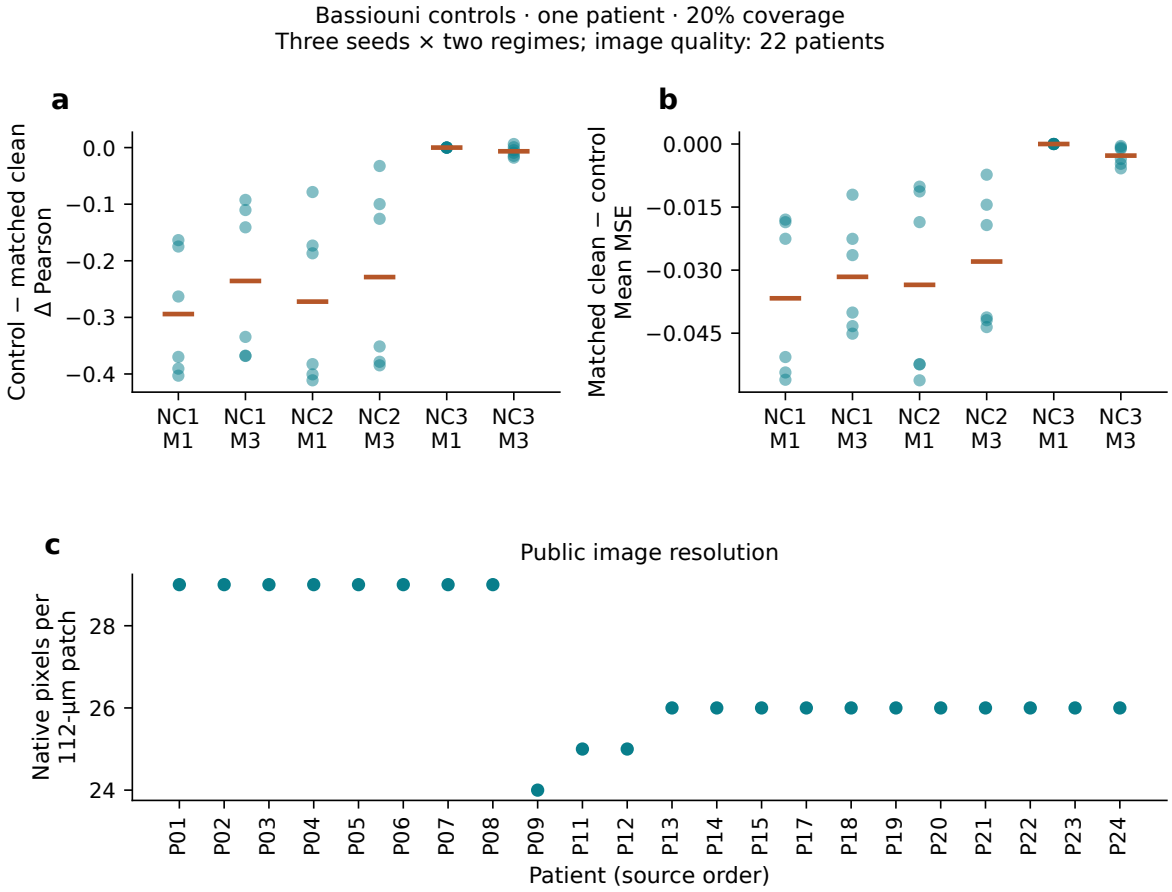


Figure S12: Replication controls and native image resolution. The independent-replication controls use the predeclared Bassiouni patient BASS_P01, 50 genes, 20% coverage, random mask seed 0, V1 geometry 0 and V2 geometry 1. **a,b,** Matched Pearson and MSE effects for M1/M3 under three control seeds, with definitions and signs as in Supplementary Fig. S11. Dots are matched control settings and bars are arithmetic means. **c,** Native pixels per nominal 112-μm patch for all 22 patients in source order. The supplied images provide only 24–29 native pixels before resizing for the encoder. This is image-input quality control, not expression-prediction performance. It explains why the image comparisons cannot establish the limits of native-resolution histopathology. Coordinate overlays for all patients are recorded in the archived qualification audit.

S3 Supplementary tables

The tables below provide compact views of the accompanying machine-readable data and definitions needed to interpret the supplementary figures. Table S2 is rounded for presentation; the digital files retain unrounded exported values. Large gene-level and distance-resolved tables are supplied as separate files rather than reproduced in full in this PDF. Mask, model and permutation seeds are computational replicates and do not increase the number of independent patients.

Table S1: Primary gene panel. The 50 genes are listed in the order fixed by variance ranking in the development-training patients. The fresh Wu and Bassiouni analyses retain all 50 genes. HER2ST analyses use the 40-gene intersection with this panel; unavailable genes are not replaced. The full data file also includes the larger development panels and cohort-specific availability records: `source_data/SuppTable1_gene_panels.csv`.

Order	Gene	Order	Gene
1	FOXA1	26	LUM
2	SCD	27	MYO5B
3	KRT7	28	SFRP4
4	ERBB2	29	KRT14
5	POSTN	30	TOP2A
6	CEACAM6	31	IL7R
7	SERHL2	32	AR
8	ANKRD30A	33	KRT5
9	EPCAM	34	PCLAF
10	ABCC11	35	KLF5
11	TACSTD2	36	SFRP1
12	GATA3	37	TRAF4
13	FASN	38	CXCR4
14	MLPH	39	MYH11
15	S100A14	40	TENT5C
16	CDH1	41	TCIM
17	CD9	42	NARS
18	KRT8	43	ACTG2
19	CCND1	44	PDK4
20	DSP	45	CD3E
21	SERPINA3	46	TRAC
22	LYPD3	47	JUP
23	TFAP2A	48	LYZ
24	TPD52	49	CENPF
25	PTGDS	50	CCDC80

Table S2: Patient-resolved reconstruction metrics. Compact view of the 25 fresh evaluation patients at 20% distributed coverage, using 50 genes. Values average mask seeds within patient. r is the median gene Pearson correlation; MSE is mean gene squared error in the common target units. I denotes image-only, ST molecular interpolation, and F fixed image fusion. Complete exported patient summaries for the available cohorts, regimes, budgets and methods are supplied in `source_data/SuppTable2_per_patient.csv`.

Patient	r_{I}	r_{ST}	r_{F}	$\mathrm{MSE}_{\mathrm{I}}$	$\mathrm{MSE}_{\mathrm{ST}}$	$\mathrm{MSE}_{\mathrm{F}}$
Wu						
1160920F	-0.045	0.359	0.335	0.846	0.082	0.095
CID4290	0.161	0.325	0.333	2.915	0.099	0.100
CID44971	0.059	0.502	0.496	2.949	0.115	0.118
Bassiouni						
BASS_P01	0.077	0.382	0.370	1.304	0.087	0.106
BASS_P02	0.009	0.459	0.431	1.067	0.093	0.127
BASS_P03	0.030	0.545	0.537	2.585	0.081	0.089
BASS_P04	-0.039	0.334	0.312	1.882	0.113	0.125
BASS_P05	0.096	0.636	0.628	1.581	0.133	0.155
BASS_P06	-0.005	0.312	0.297	1.567	0.069	0.072
BASS_P07	0.013	0.191	0.180	2.684	0.092	0.097
BASS_P08	0.095	0.403	0.393	2.120	0.088	0.095
BASS_P09	0.075	0.391	0.380	1.763	0.120	0.139
BASS_P11	-0.081	0.474	0.464	2.052	0.104	0.109
BASS_P12	0.182	0.435	0.436	1.748	0.113	0.120
BASS_P13	0.151	0.612	0.615	2.228	0.104	0.119
BASS_P14	0.041	0.540	0.536	1.458	0.126	0.154
BASS_P15	0.093	0.386	0.389	1.777	0.101	0.115
BASS_P17	0.095	0.428	0.421	1.773	0.088	0.096
BASS_P18	0.090	0.516	0.496	2.391	0.083	0.086
BASS_P19	0.068	0.537	0.501	2.431	0.068	0.078
BASS_P20	0.067	0.326	0.319	2.407	0.096	0.098
BASS_P21	0.075	0.401	0.340	2.607	0.092	0.108
BASS_P22	-0.029	0.322	0.300	2.149	0.087	0.090
BASS_P23	0.267	0.541	0.525	2.166	0.054	0.061
BASS_P24	0.106	0.350	0.302	1.626	0.048	0.052

Table S3: Per-gene results and metric definitions. The accompanying digital table, `source_data/SuppTable3_per_gene.csv.gz`, contains the exported Wu and Bassiouni gene-level results, with run, patient, panel, mask and method identifiers. Compression changes storage only. Every metric is calculated using the hidden locations scored by the corresponding protocol. The definitions below explain the gene-level statistics and their summaries; genes and repeated runs are not independent patient observations.

Quantity	Definition and interpretation
Pearson correlation	Correlation between predicted and measured values across scored locations, calculated separately for each gene. Constant targets or predictions yield undefined correlations.
Spearman correlation	Correlation between the ranks of predicted and measured values across scored locations, calculated separately for each gene. Numerical changes in tied ranks do not establish a biological improvement.
Mean squared error (MSE)	Mean squared prediction error across scored locations for each gene, evaluated in the common development-standardized log1p-count units.
Mean absolute error (MAE)	Mean absolute prediction error across scored locations for each gene, evaluated in the same target units.
Panel correlation	Median of defined gene correlations. Undefined correlations are excluded from this median and do not count as positive-correlation successes.
Panel error	Arithmetic mean of the gene-level MSEs or MAEs. All genes are retained in error summaries.
Patient and cohort summaries	Mask seeds are averaged within geometry and geometries within patient. Headline cohort values are medians of patient summaries; paired effects are calculated within patients before summarizing them.

Table S4: Negative-control families. Control definitions are experiment-specific and are given in the corresponding figure legends. Training-row controls differ from inference-time controls that permute permitted sparse inputs. Permuting anchor-expression assignments and anchor-position assignments disrupts related associations and does not constitute independent biological evidence. Exported control results are supplied in `source_data/SuppTable4_negative_controls.csv`.

Control family	Interpretation
Training-target or training-embedding row permutation	Disrupts local morphology–target pairing during development while preserving the validation and evaluation data.
Anchor-expression assignment permutation	Reassigns observed expression vectors among anchors, disrupting their correspondence with spatial position.
Anchor-position assignment permutation	Disrupts the spatial assignment of observed anchors while retaining the permitted molecular information.
Image-location or image-prediction permutation	Tests dependence on the image input while retaining the allowed molecular measurements.

Table S5: Recorded integrity and replay checks. Counts are taken from the original Wu and Bassiouni technical audits; these audits were not rerun during manuscript preparation. Method predictions and metric arrays are distinct audit units. For each prediction, hidden labels were independently replaced by NaNs and random values to test that unobserved expression could not influence model inputs. Digital records: `source_data/SuppTable5_integrity.csv`.

Audit quantity	Wu	Bassiouni
Mask trajectories	132	968
Clean run bundles	396	2,904
Control run bundles	18	18
Method prediction replays	1,656	8,766
Metric array replays	6,624	35,064
Hidden-label corruption checks	3,312	17,532

Table S6: Observation masks and sampling protocols. Coverage is the proportion of retained locations with observed complete expression vectors, not the proportion of tissue area assayed. All unobserved labels are excluded from model inputs. The three regional geometries and ten random-mask seeds are computational replicates. The accompanying `source_data/SuppTable6_masks.csv` records the exported mask identities, patients, regimes, geometries and sampling trajectories.

Protocol component	Definition
Measurement budgets	For N retained locations and $b \in \{0.10, 0.20, 0.40\}$, the observed count is $\lfloor bN + 0.5 \rfloor$. Random anchor sets use nested prefixes of the same trajectory as coverage increases.
Distributed interpolation (V1)	Anchors may be sampled throughout the retained tissue; every remaining location is scored. Predictions use up to 16 nearest observed anchors with normalized inverse-distance-squared weights.
Contiguous regional completion (V2)	A prespecified contiguous region contains the 40% of locations nearest to its center. Anchors are sampled outside that region, and only locations within it are scored. Predictions use the mean of up to 16 nearest observed anchors.
Regional geometry	Three centers are selected at the 25th, 50th and 75th percentile of first coordinate principal-component rank. The three geometries are averaged equally within patient.
Secondary spatial sampling	Label-free farthest-point sampling starts from the eligible location farthest from the eligible-coordinate centroid. It is not optimized using external expression outcomes.
Development masks	Earlier block and capture-window masks differ from the external regional-completion protocol and are interpreted separately.

Table S7: Distance-resolved evaluation and patient support. Distance is measured from each scored hidden location to its nearest observed anchor. Empty bins are omitted. Bassiouni binwise Pearson correlations require at least 20 scored locations in the corresponding run; Wu distance bins report MSE and MAE only. Error summaries may include smaller occupied bins. The patient counts below summarize the Bassiouni cohort at 20% coverage and do not define a universal distance cutoff. Exported bin-level metrics and support counts are supplied in `source_data/SuppTable7_distance_bins.csv`.

Evaluation component	Definition
Distance bins (μm)	$[0, 150)$, $[150, 300)$, $[300, 600)$, $[600, 1200)$, $[1200, 2400)$ and $[2400, \infty)$.
Bin support	Number of scored hidden locations contributing to the patient, method, mask and distance-bin summary; occupied-bin support may differ across patients and protocols.
Nearest-to-farthest comparison	Compares MSE in the nearest and farthest occupied bins within each patient.
Monotonicity	Assesses whether MSE strictly increases across the occupied distance bins.

Bassiouni patient summary	V1	V2
Higher MSE in farthest than nearest occupied bin	15/22	20/22
Strictly increasing occupied-bin MSE	6/22	13/22

Table S8: Selected supporting predictor architectures for the 50-gene panel. Dimensions denote input → hidden → output, and every listed MLP has GELU, dropout 0.1 and a linear output layer. Encoder parameters remain frozen; counts include trainable head biases. All MLPs use the shared optimizer and stopping rule in the main Methods, Experimental setup. The operators P, Q and R are defined in the main graph-representation Methods. The initial local predictor and the representation-comparison predictor are distinct fitted heads.

Predictor	Selected dimensions	Fitting settings or fixed aggregation
Initial ResNet18 ridge	$512 \to 50$	Ridge penalty 100; intercept; 25,650 coefficients.
Initial ResNet18 MLP	$512 \to 128 \to 50$	Lowest validation MSE; 72,114 head parameters.
ResNet18 mean pooling / GCN	$512 \to 128 \to 50$	PX or QX on capped Delaunay adjacency; 72,114 head parameters.
ResNet18 mean-GraphSAGE	$1024 \to 67 \to 50$	$[X, RX]$ on capped Delaunay adjacency; 72,075 head parameters.
ResNet18 representation-comparison MLP	$512 \to 256 \to 50$	Highest validation median gene Pearson; 144,178 head parameters.
Phikon local MLP	$768 \to 256 \to 50$	Same candidate grid and selection criterion; 209,714 head parameters.
Phikon spatial MLPs	$768 \to 128 \to 50$; $1536 \to 67 \to 50$	Mean pooling and mean-GraphSAGE, respectively; inherited graph and head widths.
Sparse residual correction	34 covariates plus intercept per gene	Gene-specific ridge correction of fixed fusion; penalty 10,000; 1,750 coefficients for 50 genes.

Table S9: Coverage response and contiguous-region completion in the two fresh studies. All rows use the same frozen 50-gene panel and spatial interpolation. Values are medians of patient summaries, with $n = 3$ for Wu and $n = 22$ for Bassiouni. V1 is distributed interpolation with inverse-distance-squared weighting; V2 is completion of a predefined contiguous region using the mean of nearby anchors. Both the scoring support and interpolation rule change between V1 and V2. The comparison therefore describes protocol-level deterioration rather than an isolated causal geometry effect. In both studies, every evaluation patient improved in Pearson correlation and MSE from 10% to 40% V1 coverage; every patient deteriorated in Pearson, MSE and MAE under V2 relative to V1 at 20%.

Cohort	Coverage	V1 Pearson r	V1 MSE	V2 Pearson r	V2 MSE
Wu, fresh	10%	0.313	0.1080	0.111	0.1356
	20%	0.359	0.0987	0.081	0.1366
	40%	0.393	0.0899	0.088	0.1385
Bassiouni, fresh	10%	0.380	0.1001	0.121	0.1520
	20%	0.416	0.0923	0.171	0.1496
	40%	0.465	0.0852	0.181	0.1518